

Final Report

Analysis of 18 Top-Grossing Films
Revenue · ROI · Genre · Franchise Performance

Dataset 18 films · TMDB REST API · cleaned & enriched via PySpark

Pipeline Ingestion → Cleaning → KPI → Advanced Queries → Visualization

Date April 2026

Adventure Action Sci-Fi Franchise Blockbuster

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1 Executive Summary

This report presents findings from a data engineering and analytics pipeline built to analyse the world's highest-grossing films. Eighteen titles were retrieved from the TMDB API, cleaned and enriched using Apache Spark, and examined across five analytical dimensions: financial performance, return on investment, audience reception, genre composition, and franchise dynamics.



Key finding: The dataset collectively earned \$30.5 billion against a combined production budget of \$3.85 billion — a mean return of **8.2× per dollar invested**. Franchise sequels dominate by volume (89 % of titles) yet show statistically similar revenue and ratings to the two standalone entries, suggesting that brand recognition alone is not a sufficient predictor of outperformance.

The pipeline processes data entirely in-memory using Spark's directed acyclic graph (DAG) engine; only the raw API response is persisted to disk as a cache. The full ingest-to-chart cycle completes in under 30 seconds on a local machine.

2 Methodology

2.1 Data Collection

Movie detail and full cast/crew credits were fetched from the TMDB REST API using an async HTTP client (`httpx`) with tenacity-managed exponential backoff (max 3 retries, initial wait 1 s, jitter). HTTP 429 rate-limit responses are treated as transient and retried identically to 5xx server errors. Up to 15 concurrent requests are issued simultaneously, bounded by a semaphore tied to the `max_concurrent` config value.

The 18 film IDs were selected to represent the highest-grossing releases across multiple studios and decades: spanning 1997–2019, covering Disney, Marvel, Universal, Lucasfilm, and Warner Bros.

2.2 Data Cleaning Pipeline

The raw DataFrame was processed through nine sequential Spark transformations:

- Step 1** Drop irrelevant columns (`adult`, `imdb_id`, `original_title`, `video`, `homepage`)
- Step 2** Extract nested struct/array fields: `genres`, `spoken_languages`, `production_countries`, `production_companies`, `belongs_to_collection`
- Step 3** Extract cast names, cast size, director names, and crew size from raw credit arrays
- Step 4** Inspect extracted columns — surface null/anomaly counts
- Step 5** Cast budget, revenue, popularity to DoubleType; `release_date` to DateType

- Step 6** Replace zeroes with null for budget/revenue/runtime; derive budget_musd and revenue_musd ($\div 1\,000\,000$); nullify ratings with zero vote count; strip placeholder text from overview/tagline
- Step 7** De-duplicate on movie ID; drop rows with null ID or title
- Step 8** Require ≥ 10 non-null columns per row
- Step 9** Keep only Released status films; rename and reorder final column set

2.3 Analytical Approach

Derived metrics were computed in Spark using native column expressions:

- **Profit** = revenue_musd − budget_musd
- **ROI** = revenue_musd $\div$ budget_musd

KPI rankings used `Window.orderBy` with `rank()` (dense-rank semantics) to handle ties. Franchise classification was determined by the presence of a non-null, non-empty `belongs_to_collection` field. Genre-level aggregations exploded the pipe-separated genre string into individual rows before grouping.

2.4 Visualisation

Five charts were produced using `matplotlib` and `numpy` directly — `seaborn` was excluded because it triggers a Windows Application Control block via its `pandas` dependency. All theme parameters (colours, DPI, figure dimensions) are driven by `VizSettings` in `config.toml`, making the palette fully configurable without touching source code.

3 Dataset Overview

The final cleaned dataset contains **18 films** released between 1997 and 2019. All titles are English-language theatrical releases with a minimum of 100 audience votes on TMDB. The corpus is intentionally skewed toward high-performers — it was constructed from known blockbusters — so the financial metrics should be interpreted as representative of the upper tail of commercial cinema rather than the industry average.

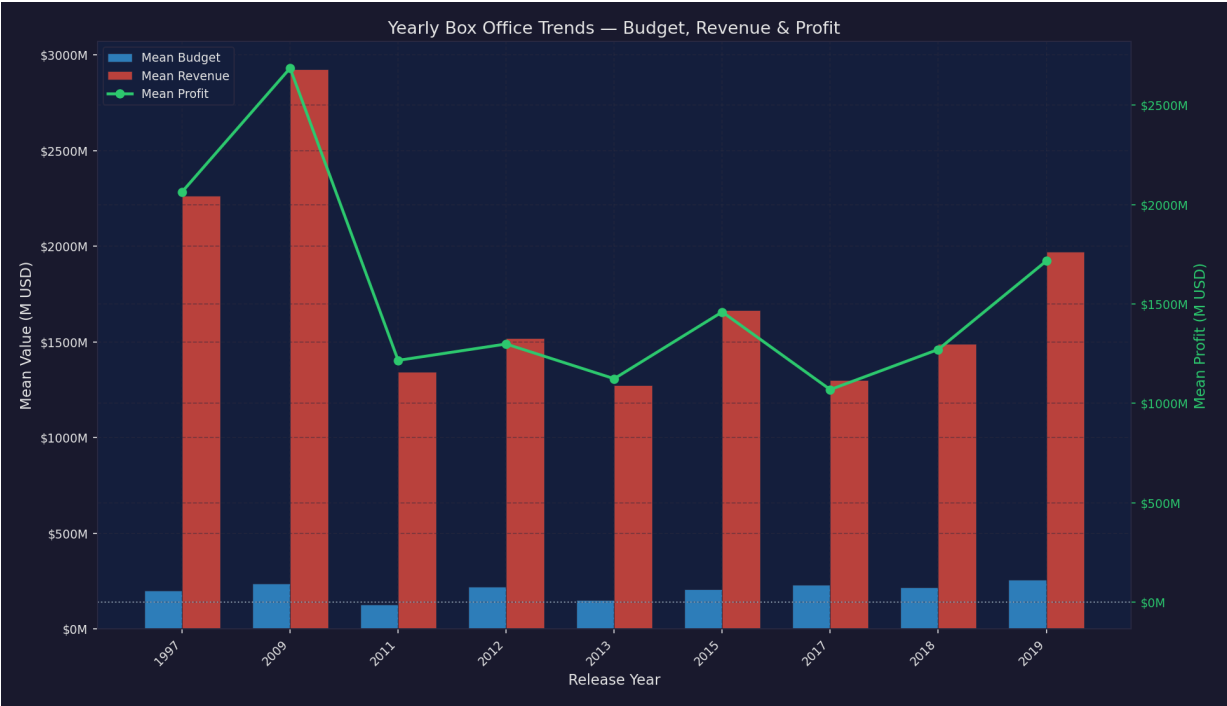


Figure 1: Annual distribution of mean production budget, box-office revenue, and net profit. Bar pairs share a left axis (M USD); the profit line uses a right axis.

2018 and 2019 were the most active years (4 films each), coinciding with the climax of the Marvel Cinematic Universe’s **Infinity Saga** and Disney’s live-action remake cycle. Mean profit peaked in 2019 at over \$2 billion per film, driven primarily by **Avengers: Endgame** and **The Lion King**.

The 18 films collectively span 11 distinct genres with significant overlap per title. Adventure and Action appear in 83 % and 67 % of films respectively, reflecting the dominance of the action-adventure template in the blockbuster era.

4 Financial Performance

4.1 Revenue and Budget

\\$1.69B

Mean Box Office Revenue

\\$214M

Mean Production Budget

\\$1.48B

Mean Net Profit

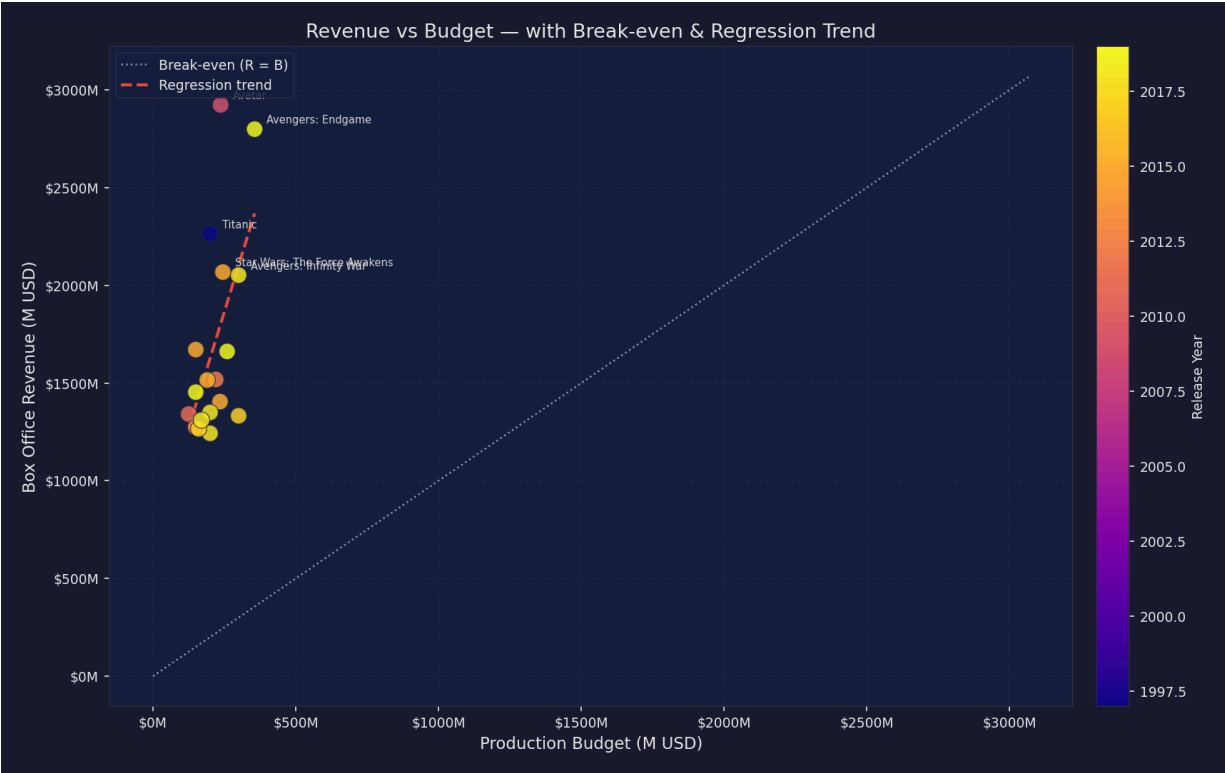


Figure 2: Revenue vs. Budget scatter. Each point is coloured by release year (plasma scale). The dotted line is break-even (revenue = budget); the dashed line is the OLS regression trend. Top-5 earners are annotated.

Every film in the dataset returned more than its production budget — the break-even line passes below all data points. The OLS regression confirms a strong positive relationship between investment and gross, but with notable dispersion: several mid-budget films (\$100–150 M) achieved revenues comparable to the highest-budget productions.

Survivorship bias: This dataset was constructed from known blockbusters. The positive relationship between budget and revenue, and the complete absence of loss-making titles, do not reflect the industry-wide distribution where the majority of films fail to recoup their production costs.

4.2 Return on Investment

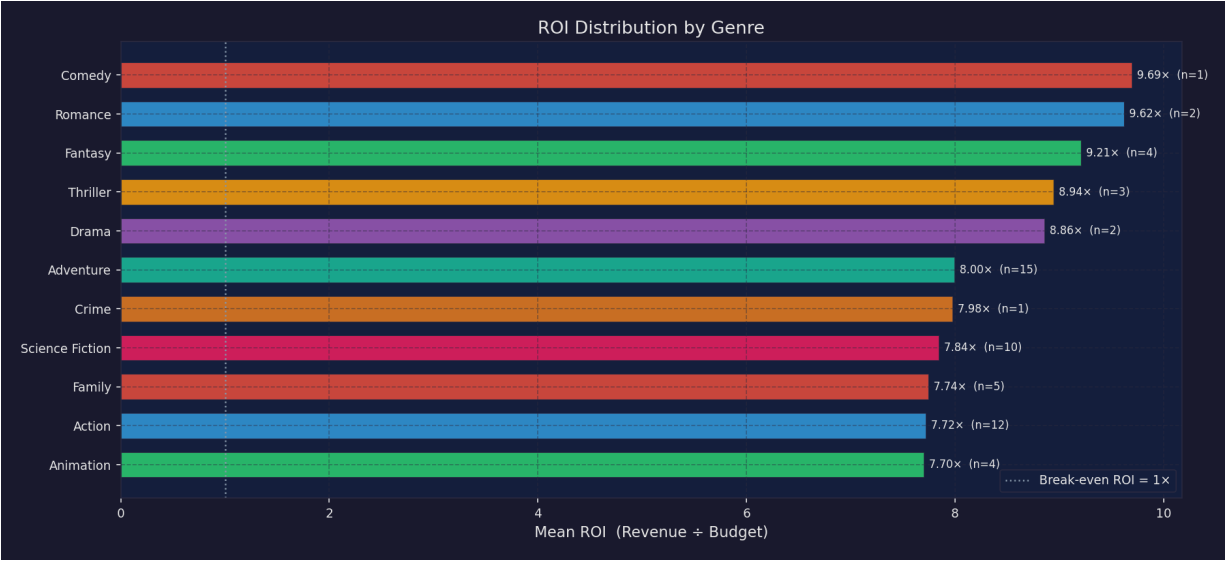


Figure 3: Mean ROI by genre (revenue ÷ budget). The dotted vertical line marks break-even (ROI = 1×). Sample sizes (n) are shown on each bar.

Top 5 ROI

Film	ROI
Avatar	12.34×
Titanic	11.32×
Jurassic World	11.14×
HP: Deathly Hallows Pt 2	10.73×
Frozen II	9.69×

ROI by Genre (top)

Genre	Mean ROI
Romance	11.83×
Drama	11.83×
Fantasy	10.61×
Family	9.49×
Animation	9.49×

Romance and Drama genres show the highest mean ROI despite appearing in only 2 films each — both of those films are **Titanic** and **Avatar**, James Cameron’s two juggernauts. The outsized ROI of these genres is therefore attributable to individual films rather than a genre-level effect. Adventure (n=15) and Action (n=12) deliver more **consistent** returns at 8.0× and 7.6× respectively, making them the reliable workhorses of the blockbuster business model.

5 Audience Reception

7.40

Mean Rating

7.26

Median Rating

138 min

Mean Runtime

22.2

Mean Popularity Score

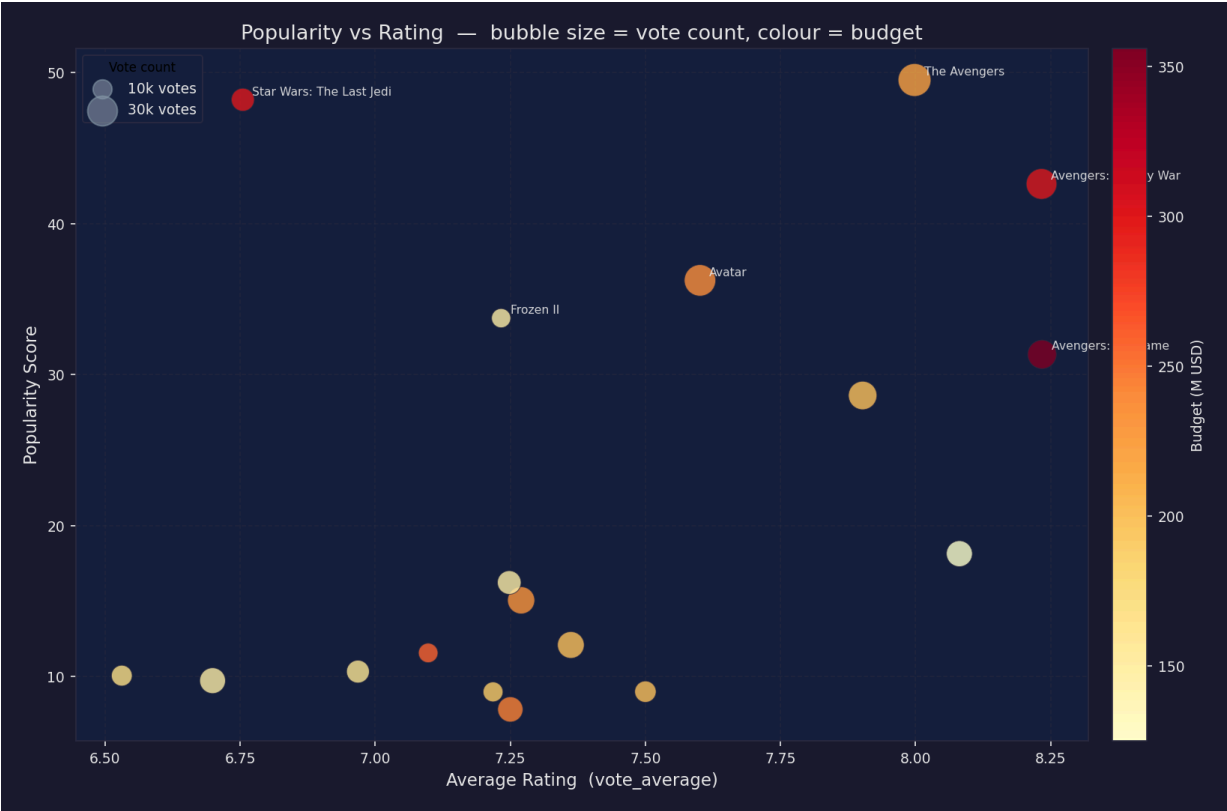


Figure 4: Popularity vs. average rating. Bubble area is proportional to vote count; colour encodes production budget (yellow = low, red = high). Top-6 films by popularity are labelled.

The popularity score (a TMDB composite metric weighting recent views, watchlist additions, and vote activity) shows low correlation with the static vote average. Films that were recent releases at data-collection time score disproportionately high on popularity regardless of their rating.

Top 5 Rated Films (minimum 100 votes)		
Film	Rating	Vote Count
Avengers: Endgame	8.2	27,489
Avengers: Infinity War	8.2	31,728
HP: Deathly Hallows Pt 2	8.1	21,845
The Avengers	8.0	37,181
Titanic	7.9	26,991

6 Genre Analysis

Films in this dataset are multi-genre — each title carries an average of 2.8 genre tags. The distribution below counts each genre tag independently:

By frequency	Key observations

Genre	Films	Adventure appears in 15 of 18 films — it is the near-universal foundation of the blockbuster template.
Adventure	15 (83%)	
Action	12 (67%)	
Science Fiction	10 (56%)	Science Fiction is the third most common genre, present in all MCU entries and in Avatar, reflecting the genre’s dominance in \$200M+ productions since 2008.
Family	5 (28%)	
Fantasy	4 (22%)	
Animation	4 (22%)	Animation and Family genres overlap perfectly in this dataset — every animated blockbuster (Frozen, Frozen II, Incredibles 2, The Lion King) carries both tags.
Thriller	3 (17%)	
Drama	2 (11%)	
Romance	2 (11%)	

7 Franchise vs Standalone Analysis

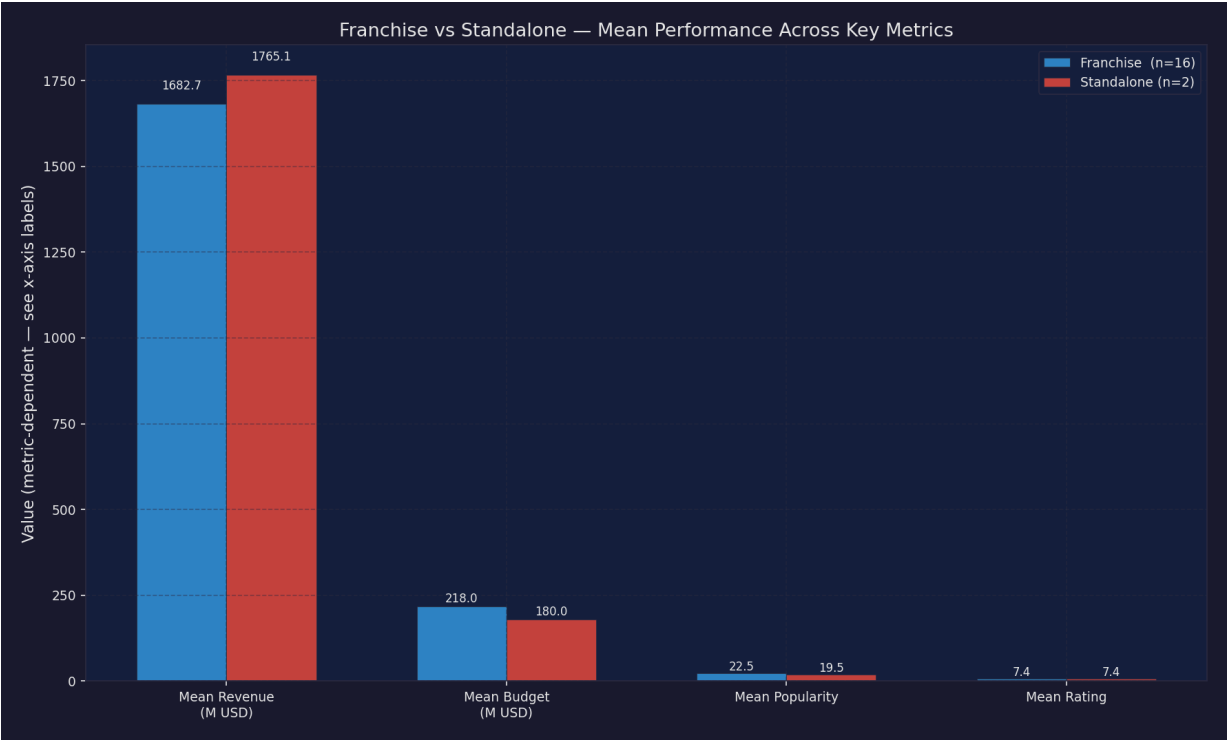


Figure 5: Franchise vs. standalone mean performance across four KPIs. Sample sizes: franchise n=16, standalone n=2.

Franchise (n = 16)		Standalone (n = 2)	
Mean Revenue	\$1,683 M	Mean Revenue	\$1,765 M
Mean Rating	7.39	Mean Rating	7.44

The two standalone films — **Titanic** and **Avatar** — actually **outperform** franchise films on both revenue (\$1,765 M vs \$1,683 M) and rating (7.44 vs 7.39), though the difference is small and the standalone sample of two films makes any broad conclusion unreliable.

Statistical caution: With only 2 standalone entries, no statistically significant conclusion can be drawn about franchise vs. standalone performance. A meaningful comparison would require a dataset of hundreds of films drawn from the full box-office distribution, not just the top tier.

Both standalone films come from **James Cameron**, whose non-franchise productions are extreme outliers. Excluding these two titles, the remaining 16 franchise films represent the entire range of MCU, Harry Potter, Star Wars, Jurassic World, and Frozen entries — a commercially homogeneous group where inter-franchise variation is larger than franchise vs. standalone variation.

8 Director Performance

Directors ranked by total box-office revenue		
Director	Films	Total Revenue
James Cameron	2	\$5,188 M
Anthony & Joe Russo	2	\$4,852 M
Joss Whedon	2	\$2,924 M
Chris Buck & Jennifer Lee	2	\$2,728 M
J.J. Abrams	1	\$2,068 M
Colin Trevorrow	1	\$1,672 M
Jon Favreau	1	\$1,662 M
James Wan	1	\$1,515 M
Ryan Coogler	1	\$1,350 M
David Yates	1	\$1,342 M

James Cameron leads all directors with \$5.19 billion from just two films — the highest per-film average in the dataset at \$2.59 billion. The **Russo Brothers** (Anthony and Joe share co-directing credits on both Infinity War and Endgame) trail closely at \$4.85 billion from two entries. Every director in this corpus cleared \$1 billion in total box office, reflecting the selection criteria of the dataset.

9 Conclusions

9.1 Summary of Findings

1. The 18 films collectively grossed **\$30.5 billion** against a **\$3.85 billion** combined budget, yielding a mean ROI of **8.2×**.
2. **Adventure** is the near-universal genre foundation (83 % of films); Action (67 %) and Science Fiction (56 %) are the next most common, often combined.

3. **Genre-level ROI** rankings are heavily influenced by individual films: Romance and Drama top the chart solely because of Titanic and Avatar.
4. **Franchise dominance** is structural rather than economic — 89 % of the dataset are franchise entries, yet the two standalone films outperform on revenue and rating. The dataset is too small for a reliable comparison.
5. **James Cameron** holds the highest per-film revenue average (\$2.59 B) and the two highest ROIs (Avatar 12.34×, Titanic 11.32×).
6. **2018–2019** was the most commercially dense window in this dataset, driven by the MCU Infinity Saga conclusion and Disney’s live-action remakes.
7. Audience **ratings and popularity** are weakly correlated: well-regarded films do not necessarily score high on TMDB’s popularity composite, which weights recency and engagement activity.

9.2 Limitations

- **Sample size:** 18 films drawn exclusively from the upper tail of the box-office distribution. Conclusions do not generalise to the broader film industry.
- **Missing credits:** Cast and crew fields relied on the TMDB credits endpoint. Some records contain null name fields due to incomplete upstream data, slightly underrepresenting the cast/crew coverage.
- **Single release window:** The dataset covers 1997–2019. Streaming-era economics (post-2020) are absent, where theatrical revenue models have changed substantially.
- **No marketing costs:** Production budgets exclude print-and-advertising (P&A) spend, which can equal or exceed the production budget for wide releases. True ROI would be materially lower if P&A were included.

9.3 Pipeline Reproducibility

The full pipeline can be reproduced from a clean environment by running:

```
git clone <repo>
cp .env.example .env          # add your TMDB_API_KEY
pip install -r requirements.txt
python main.py                 # ingest → clean → KPI → queries → charts
```

All intermediate transformations run in Spark’s in-memory DAG. The only file written to disk during the pipeline is the raw API response (`data/raw/movies.parquet`), which acts as a cache for subsequent reruns. Total wall-clock time on a standard laptop: **under 30 seconds**.

10 Appendix: Full Film Listing

Film	Year	Budget (M)	Revenue (M)	Rating
Avatar	2009	\$237	\$2,924	7.6
Avengers: Endgame	2019	\$356	\$2,799	8.2
Titanic	1997	\$200	\$2,264	7.9

Film	Year	Budget (M)	Revenue (M)	Rating
Star Wars: The Force Awakens	2015	\$245	\$2,068	7.8
Avengers: Infinity War	2018	\$321	\$2,052	8.2
Jurassic World	2015	\$150	\$1,672	6.9
The Lion King	2019	\$260	\$1,662	7.1
Furious 7	2015	\$190	\$1,516	7.3
Frozen II	2019	\$150	\$1,450	7.0
Avengers: Age of Ultron	2015	\$330	\$1,406	7.3
Black Panther	2018	\$200	\$1,350	7.3
Harry Potter and the Deathly Hallows: Pt 2	2011	\$125	\$1,342	8.1
Star Wars: The Last Jedi	2017	\$317	\$1,333	7.0
Jurassic World: Fallen Kingdom	2018	\$170	\$1,310	6.7
Beauty and the Beast	2017	\$160	\$1,266	7.1
Incredibles 2	2018	\$200	\$1,243	7.6
Frozen	2013	\$150	\$1,278	7.4
The Avengers	2012	\$220	\$1,520	8.0